

Anup Rai

+977-9827146720 | raianup806@gmail.com | linkedin.com/in/anup-rai-095695343 | github.com/Anup806 | anuprai1.com.np

SUMMARY

Electronics and Communication Engineering student focused on machine learning, data analytics, and LLM-based systems. Hands-on experience building end-to-end intelligent applications, including retrieval-augmented generation (RAG) pipelines, local LLM-powered question answering systems, and real-time computer vision interfaces. Strong foundation in Python, Scikit-learn, and data processing with practical exposure to embedding-based retrieval and context-aware AI systems. IEEE Best Paper Award recipient (AI ML Track, 2026), with a demonstrated ability to translate research into deployable solutions.

TECHNICAL SKILLS

- Programming Languages:** Python, C, C++
- Machine Learning:** Supervised/Unsupervised Learning, K-Means, Scikit-learn, NumPy, Pandas
- Probabilistic Models:** Bayesian Knowledge Tracing (BKT), Deep Knowledge Tracing (DKT)
- Computer Vision:** OpenCV, MediaPipe
- Tools & Platforms:** Git, Jupyter, Google Colab, Unity, Ollama
- Other:** Graphic Design, Video Editing

PROJECTS

- Inclusive Mixed Reality Math Experience Using ML-Based Models** 2025 – 2026
Final Year Project Python, OpenCV, MediaPipe, Scikit-learn
- Built a real-time gesture-controlled tutoring system integrating computer vision and adaptive ML
 - Implemented Bayesian Knowledge Tracing (BKT) for adaptive mastery modeling
 - Applied clustering techniques to enable personalized learning pathways
- MediaPipe Cube Control via Hand Gesture** 2026
Computer Vision / HCI Unity, C#, MediaPipe
- Developed a low-latency real-time hand-tracking interface using MediaPipe integrated with Unity
 - Designed modular gesture-state architecture for scalable gesture recognition

RESEARCH

- DKT-Based AR Framework for Predictive Learning Assessment** 2025 – 2026
IEEE ICTP 2026 | Best Paper Award – AI & ML Track
- Designed an adaptive Augmented Reality system using Deep Knowledge Tracing (DKT)
 - Developed system architecture following Research-Through-Design (RTD) methodology
 - Achieved Accuracy: 0.84 and PR-AUC: 0.82 in real-world classroom deployment

EDUCATION

- Nepal Engineering College, Pokhara University** Bhaktapur, Nepal
B.E. in Electronics & Communication Engineering 2021 – 2026
- Pokhara Aadarsha Secondary School** Pokhara, Nepal
Science (Grade 11 – 12) 2019 – 2021

CERTIFICATIONS & HONORS

- **Best Paper Award** — IEEE ICTP 2026 (AI & ML Track)
- Data Science with Python — Code IT
- IEEE Student Volunteer